08: AXI4 Lite

Engr 315: Hardware / Software Codesign

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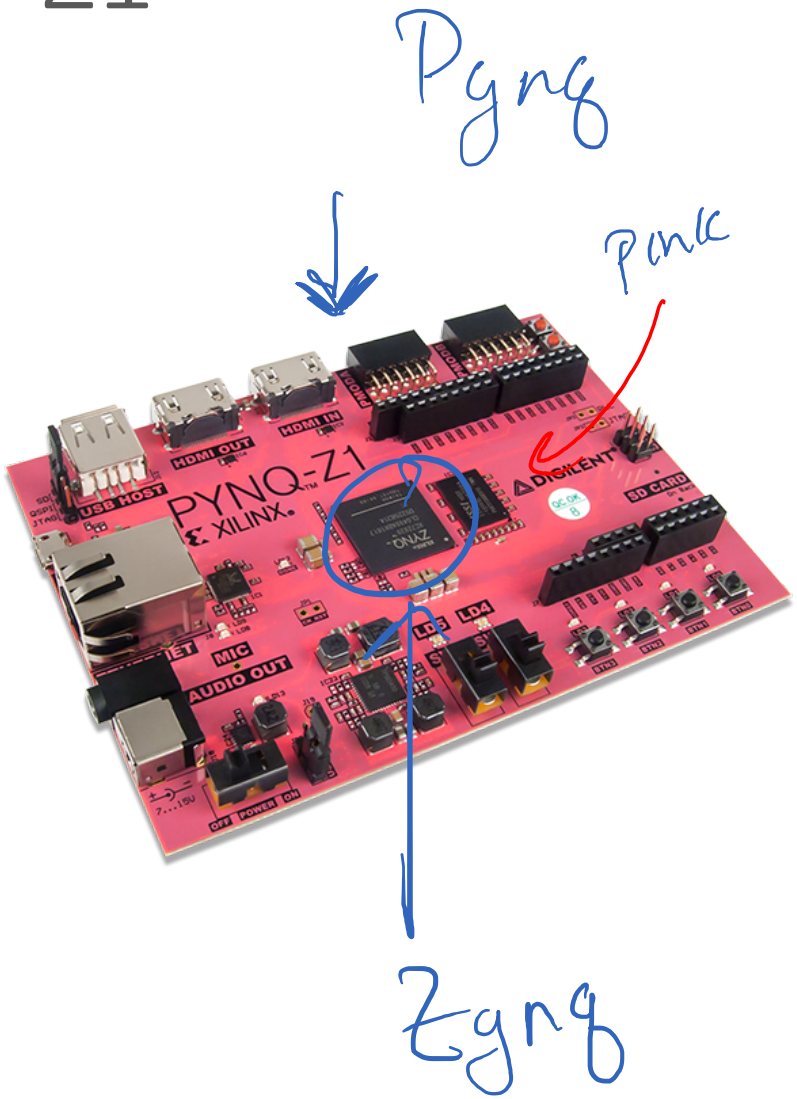
Announcements

- P2 is due Friday
- Labs/Office Hours will continue to be virtual for now.
- Working on hardware box. Updates next week.

For P3+ we'll be using the Pynq-Z1

- System-on-Chip
  - SoC - "S-O-C" or "Sock"
- Contains both FPGA and CPU 2x
- Ubuntu
- Runs Linux + Python
- Interface is a Jupyter Notebook



Hardware

- We're going to make hardware available for you.
- You can work @ home, or in Luddy 4111.

more details on hardware
Next week

Luddy 4111

- Passcode: 2-3-5
- Machines have Vivado pre-installed.
- Also have ethernet cable. Use it to connect to the Pynq.

Pure Remote

Working from home

- We will try to help where we can
- You will need your Pynq networked

→ USB → Ethernet Adapters

https://engr315.github.io

Quick Links

Syllabus

Lecture Slides

Other Downloads

Autograder

Canvas ***(Registered students only)***

Zoom ***(Requires students only)***

- Lecture
- Labs / Office Hours

Slack

Remote Setup

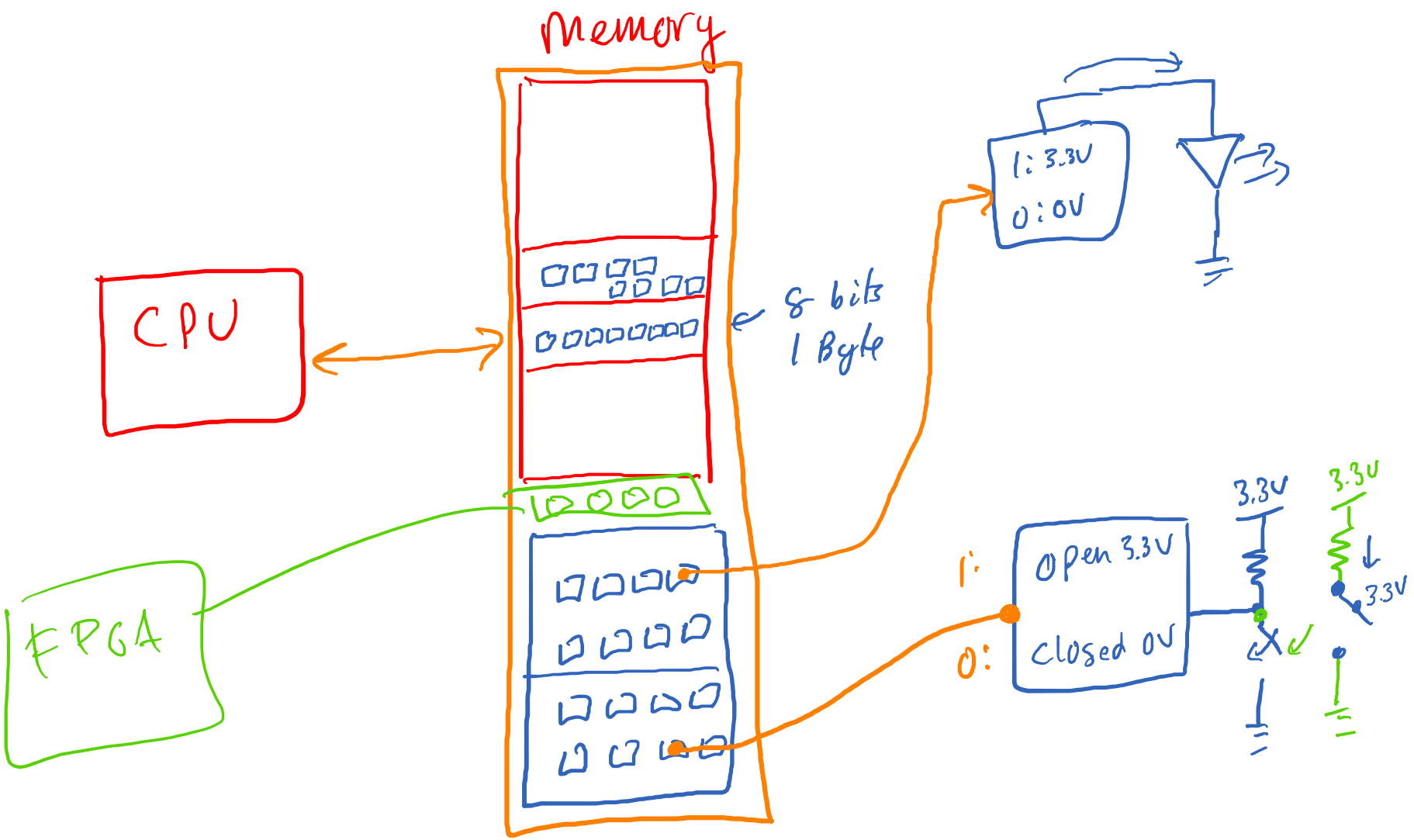
Pynq Network Setup

Optimizations thus far

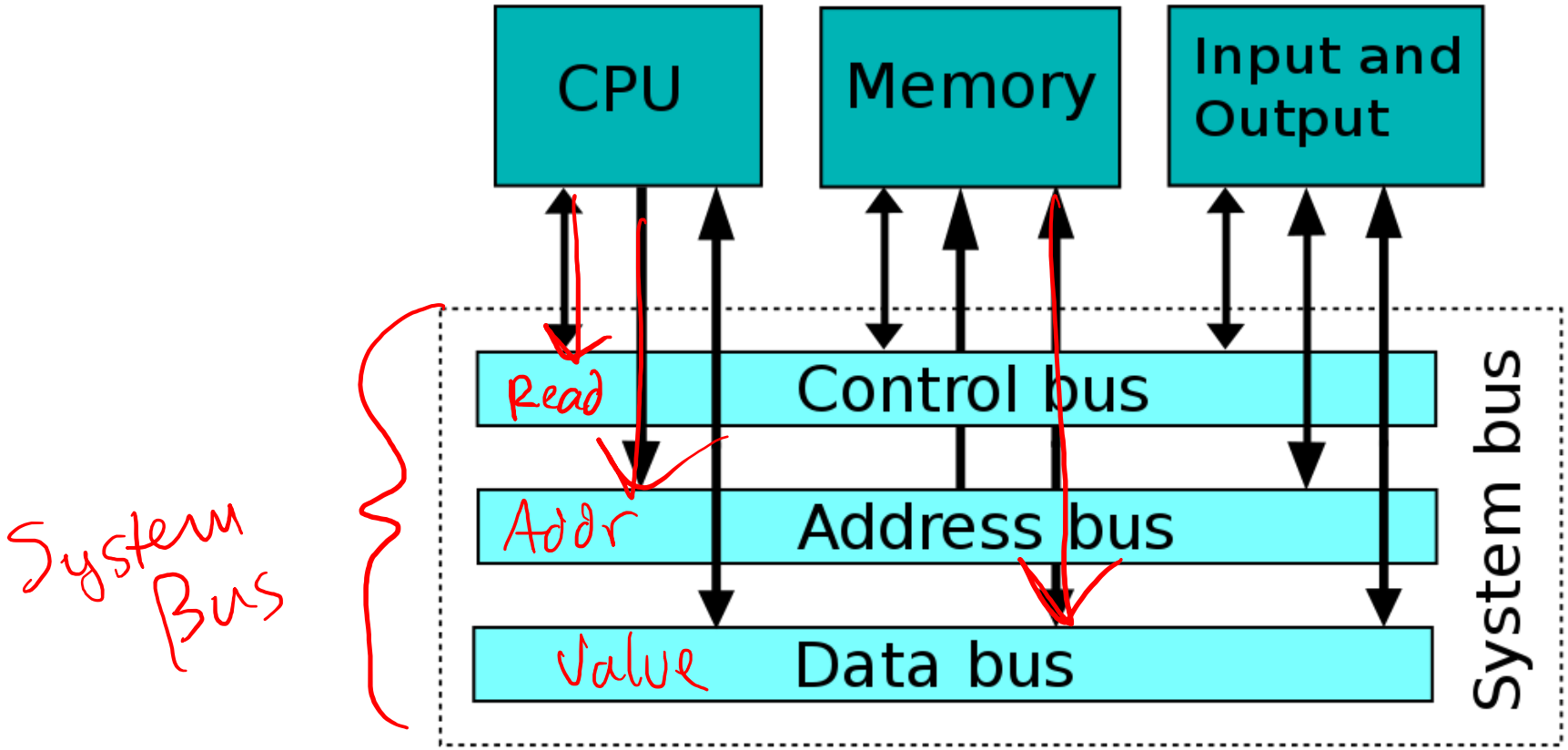
Software

- Algorithmic complexity
- Removing redundant computation
- Multithreading
- Multiprocessing*
- Python/C/Asm Interfacing
- Map to Hardware

Review: Memory-Mapped I/O



Review: The System Bus



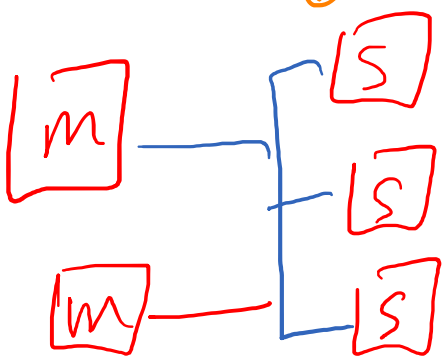
ARM AXI Bus

- “Advanced eXtensible Interface” Bus Version 4, “AXI4”

ARM AXI Bus

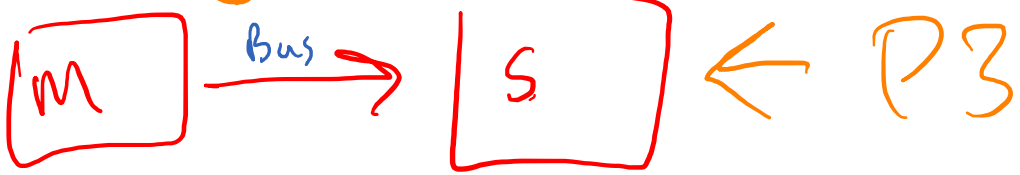
- "Advanced eXtensible Interface" Bus Version 4, "AXI4"
- Three Variants
  - AXI4: Fast but complicated; Memory-mapped
  - AXI4 Lite: Slow but simple; Memory-mapped
  - AXI4 Stream: Fast and simple; Not memory-mapped
    - P3 uses this

Mem-Mapped



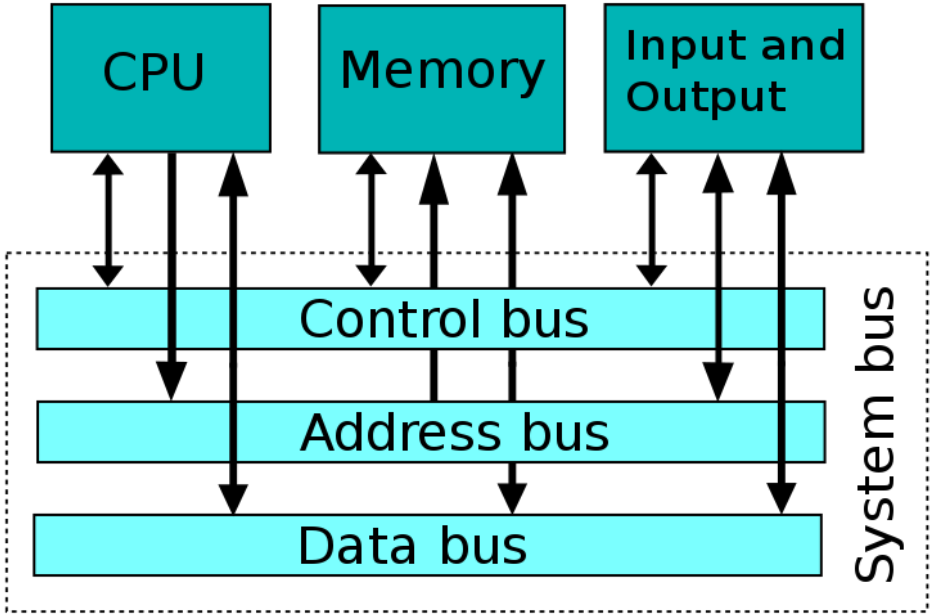
P4

Not Mem Mapped



P3

Why AXI4 Lite?

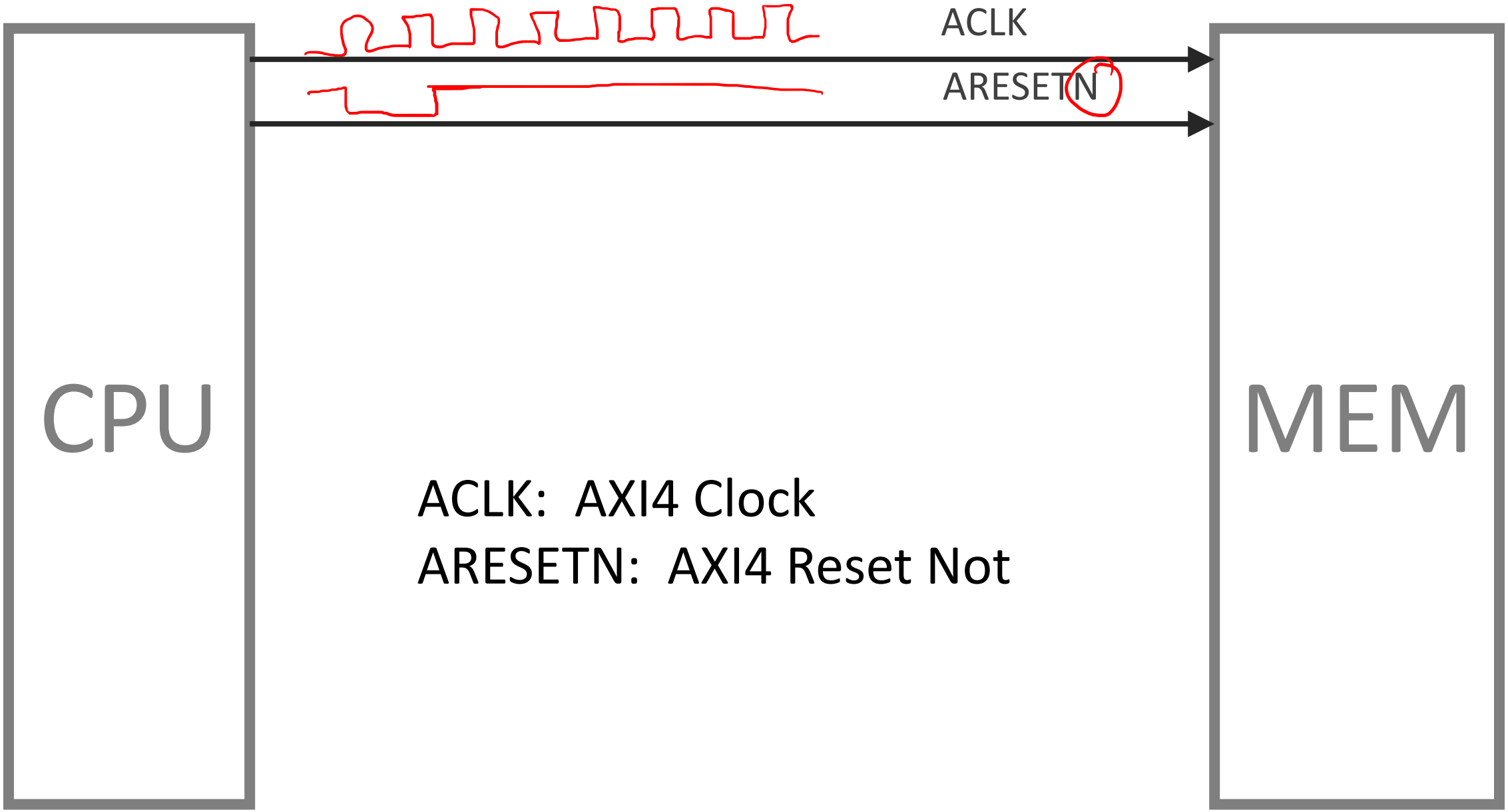


Xilinx AXI Reference Guide:

"AXI4-Lite is a light-weight, single transaction memory mapped interface. It has a small logic footprint and is a simple interface to work with both in design and usage. "

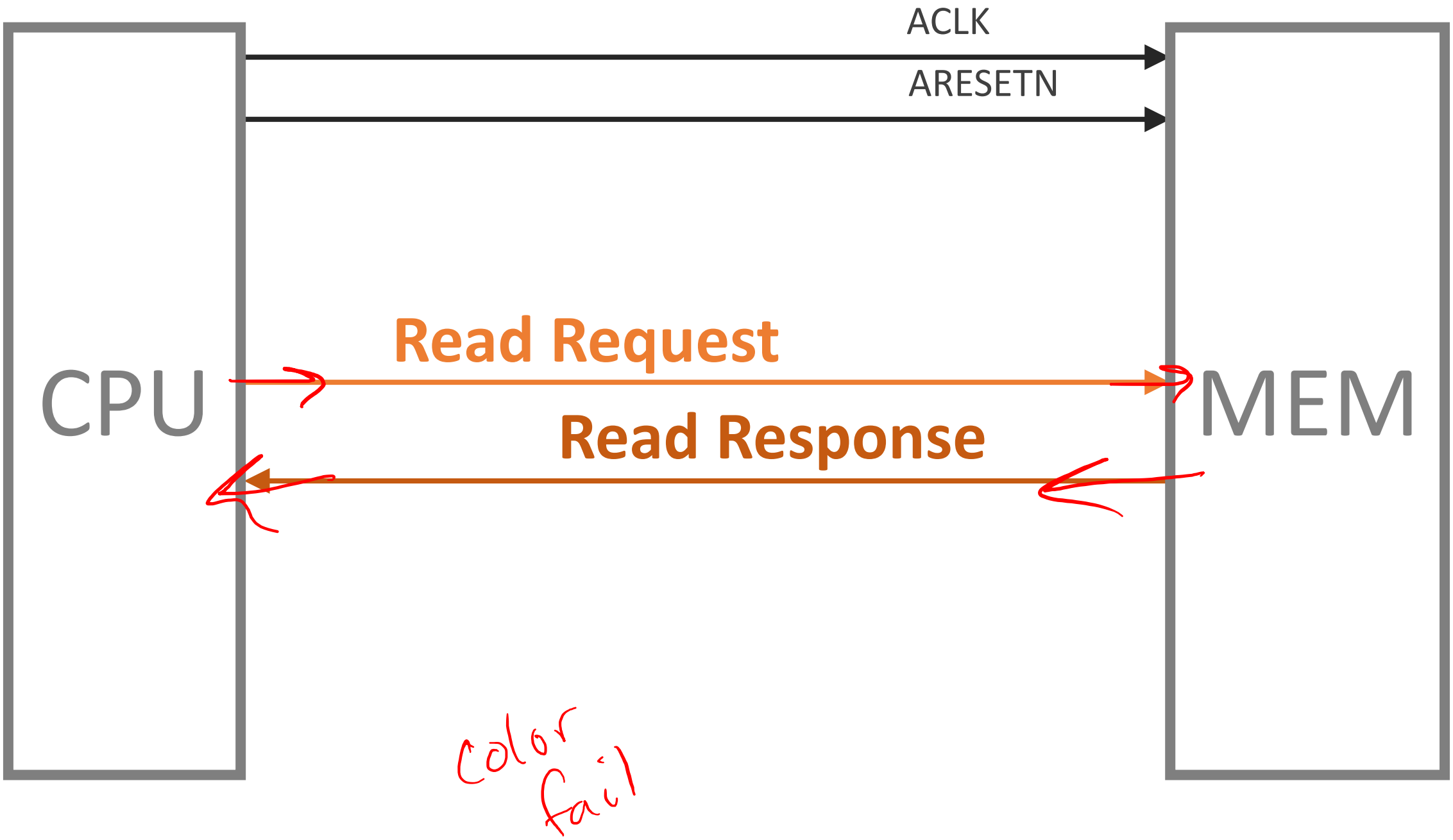
CPU

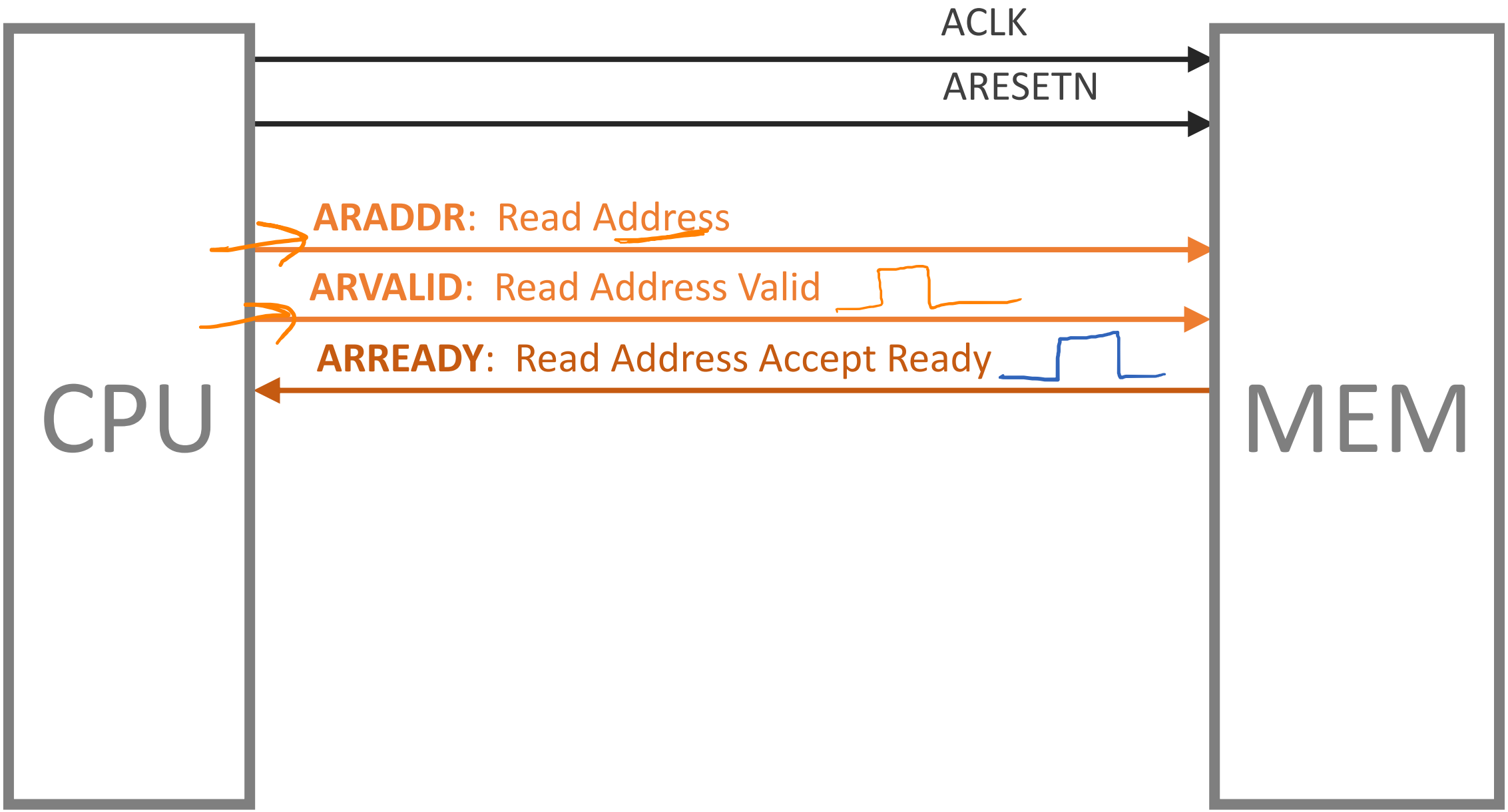
MEM



ACLK: AXI4 Clock

ARESETN: AXI4 Reset Not


ACLK
ARESETN
CPU
Read Request
Read Response
MEM
color
fail


ACLK
ARESETN
ARADDR: Read Address
ARVALID: Read Address Valid
ARREADY: Read Address Accept Ready
CPU
MEM

AXI4 Handshaking

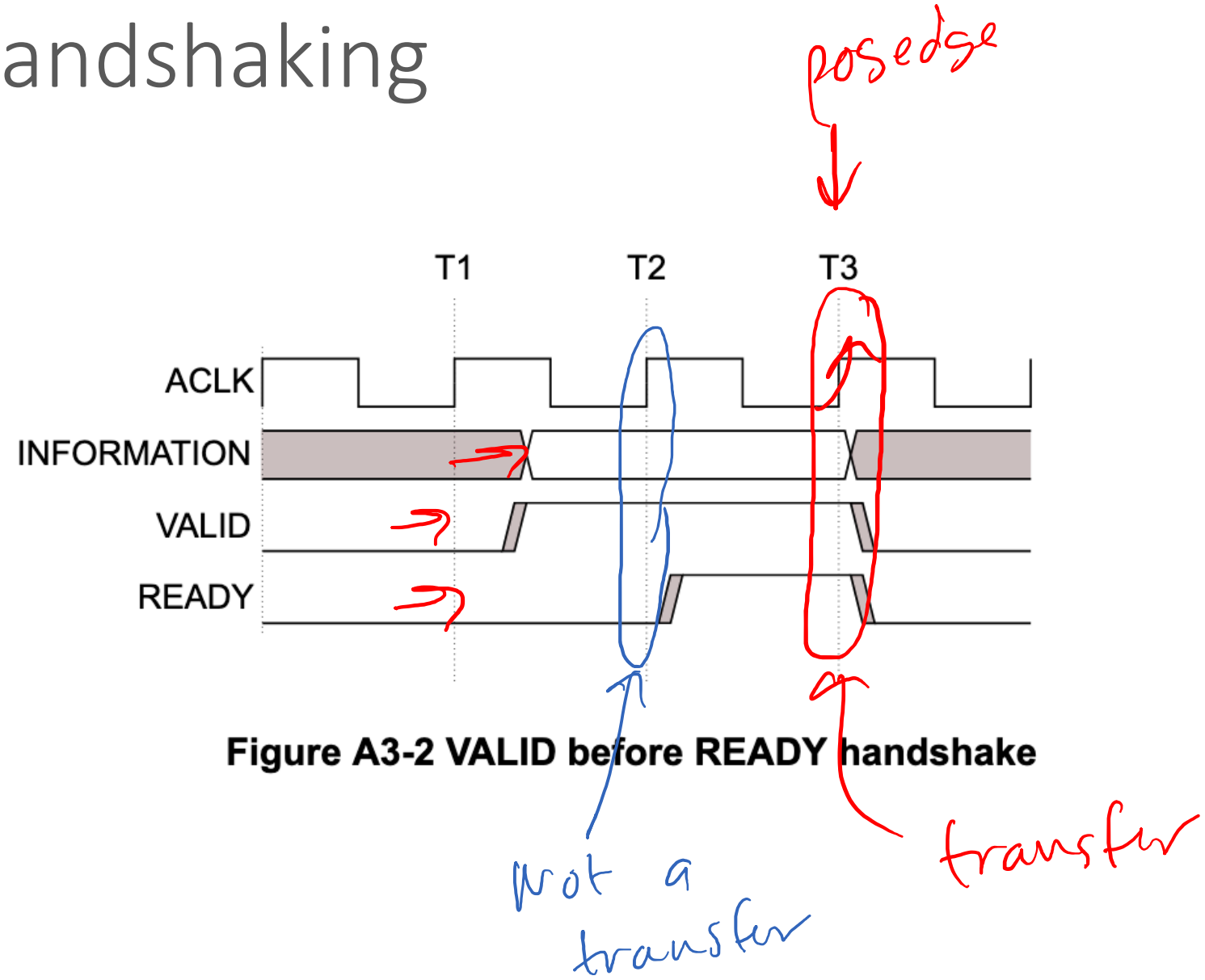
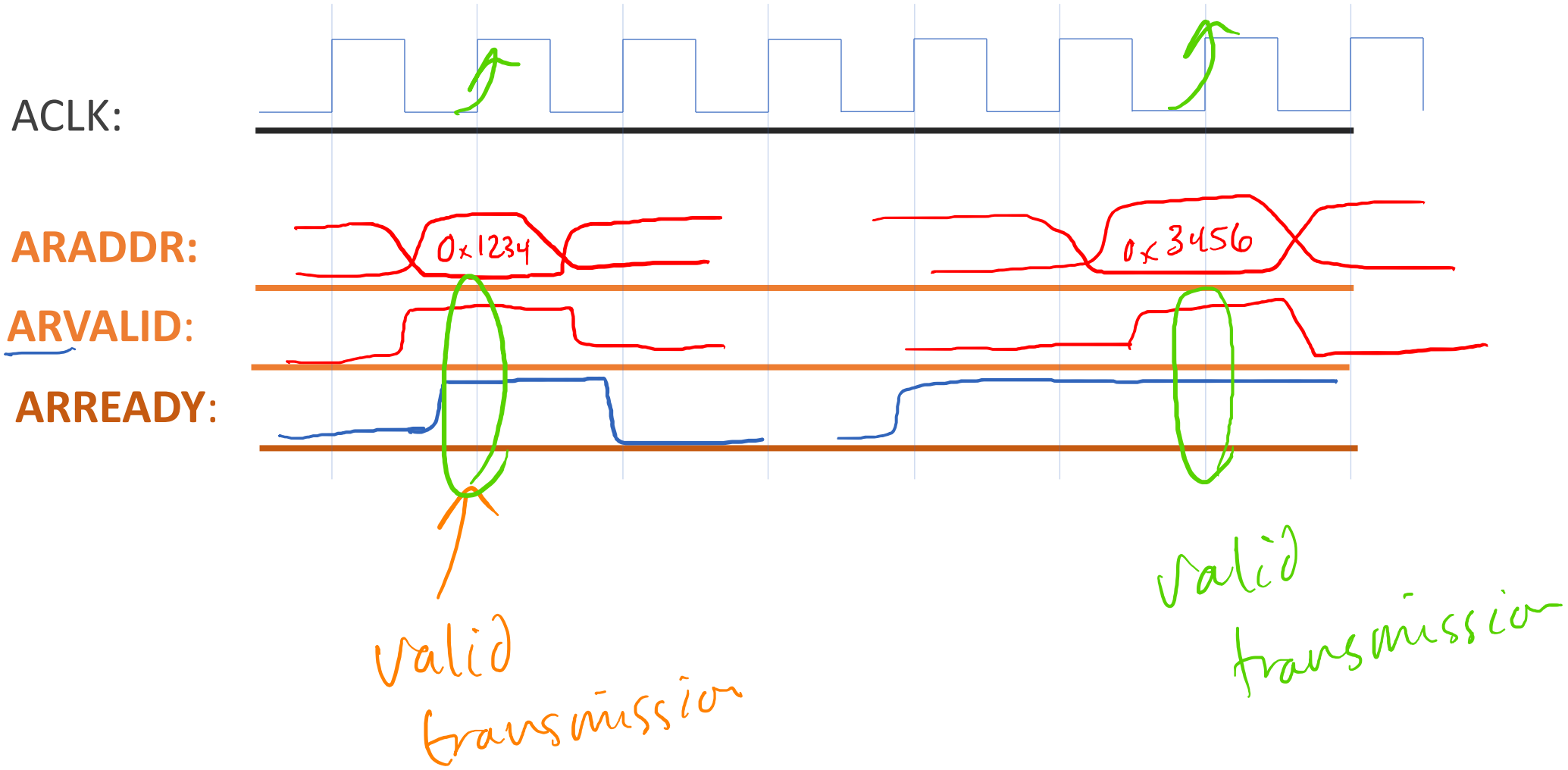
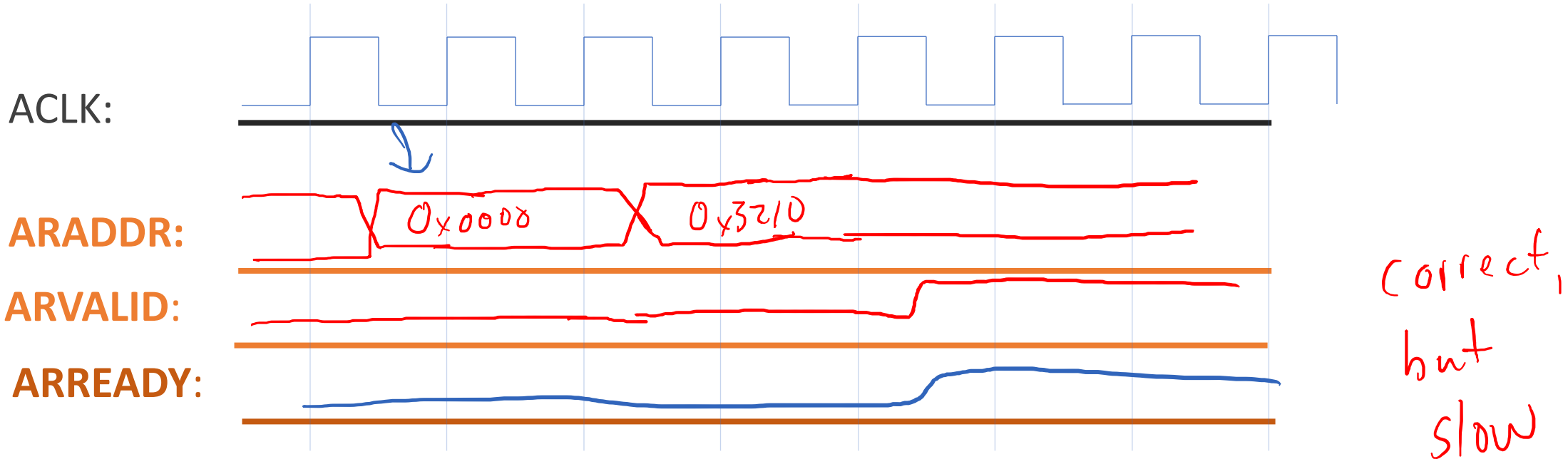


Figure A3-2 VALID before READY handshake

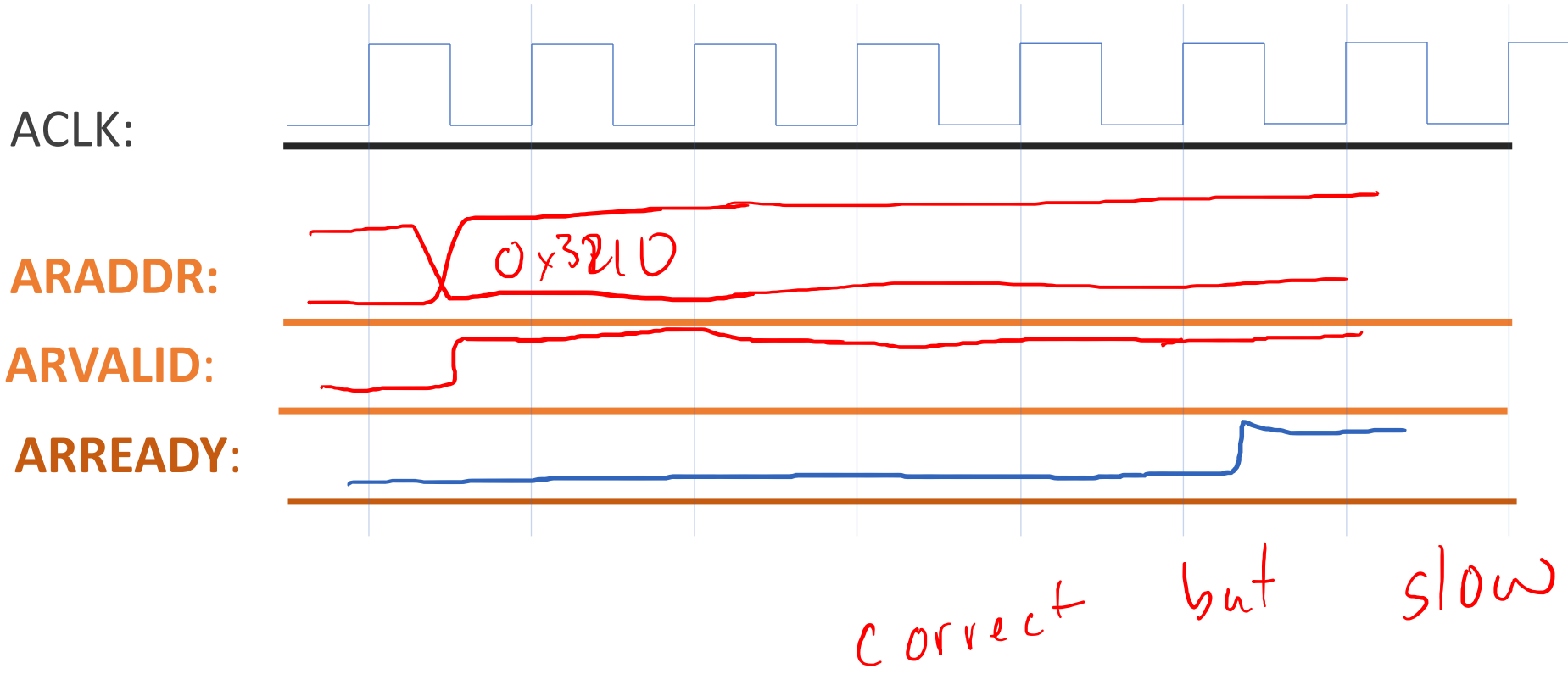
AXI4 Lite Read Transaction

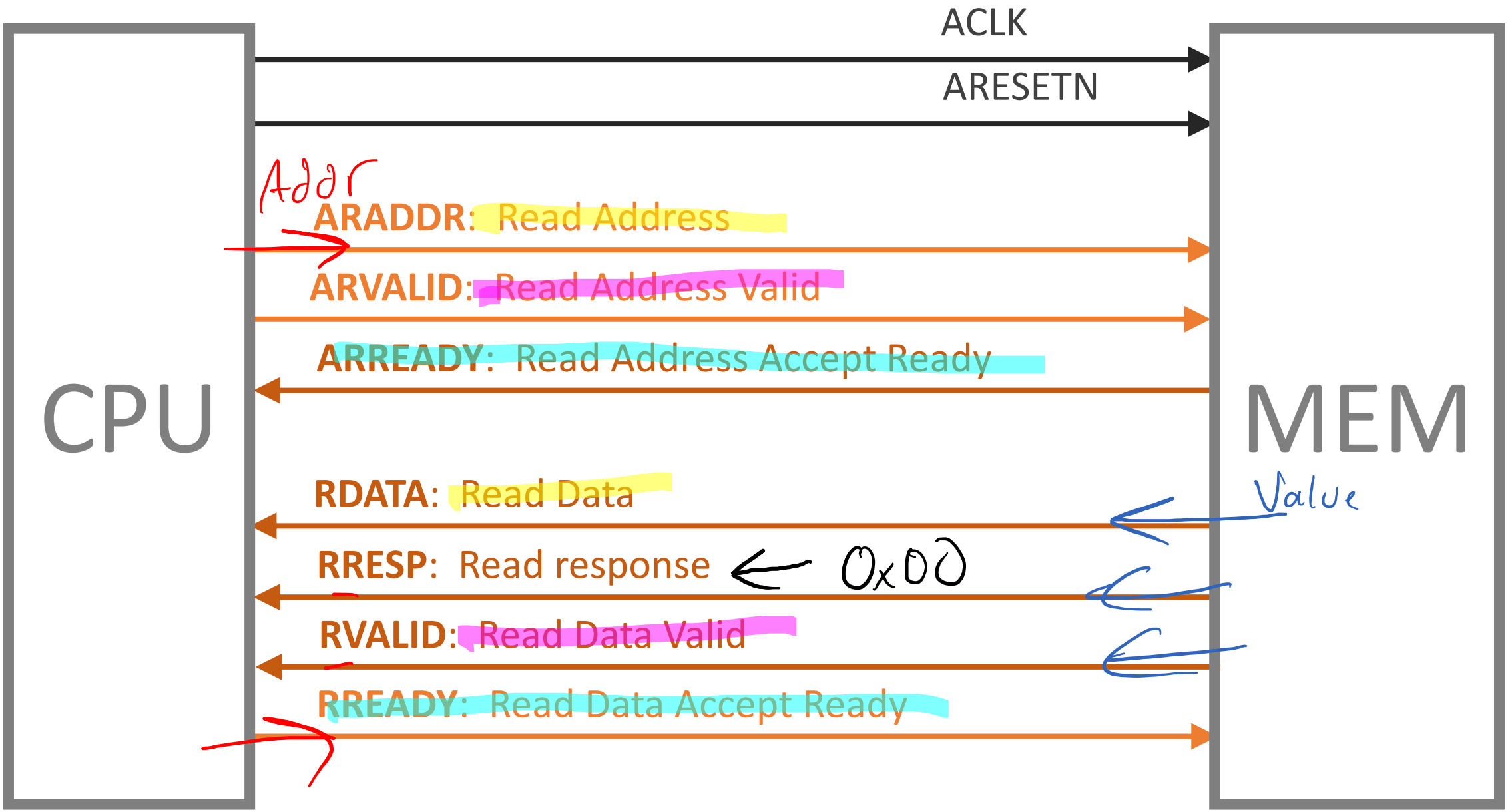


What if?


ACLK:
ARADDR:
0x0008
0x3210
ARVALID:
ARREADY:
correct, but slow

What if?




ACLK
ARESETN
Addr
ARADDR: Read Address
ARVALID: Read Address Valid
ARREADY: Read Address Accept Ready
CPU
MEM
RDATA: Read Data
Value
RRESP: Read response
0x00
RVALID: Read Data Valid
RREADY: Read Data Accept Ready

What is RRESP?

Table A3-4 RRESP and BRESP encoding

RRESP[1:0] BRESP[1:0]	Response
0b00	OKAY
0b01	EXOKAY
0b10	SLVERR
0b11	DECERR

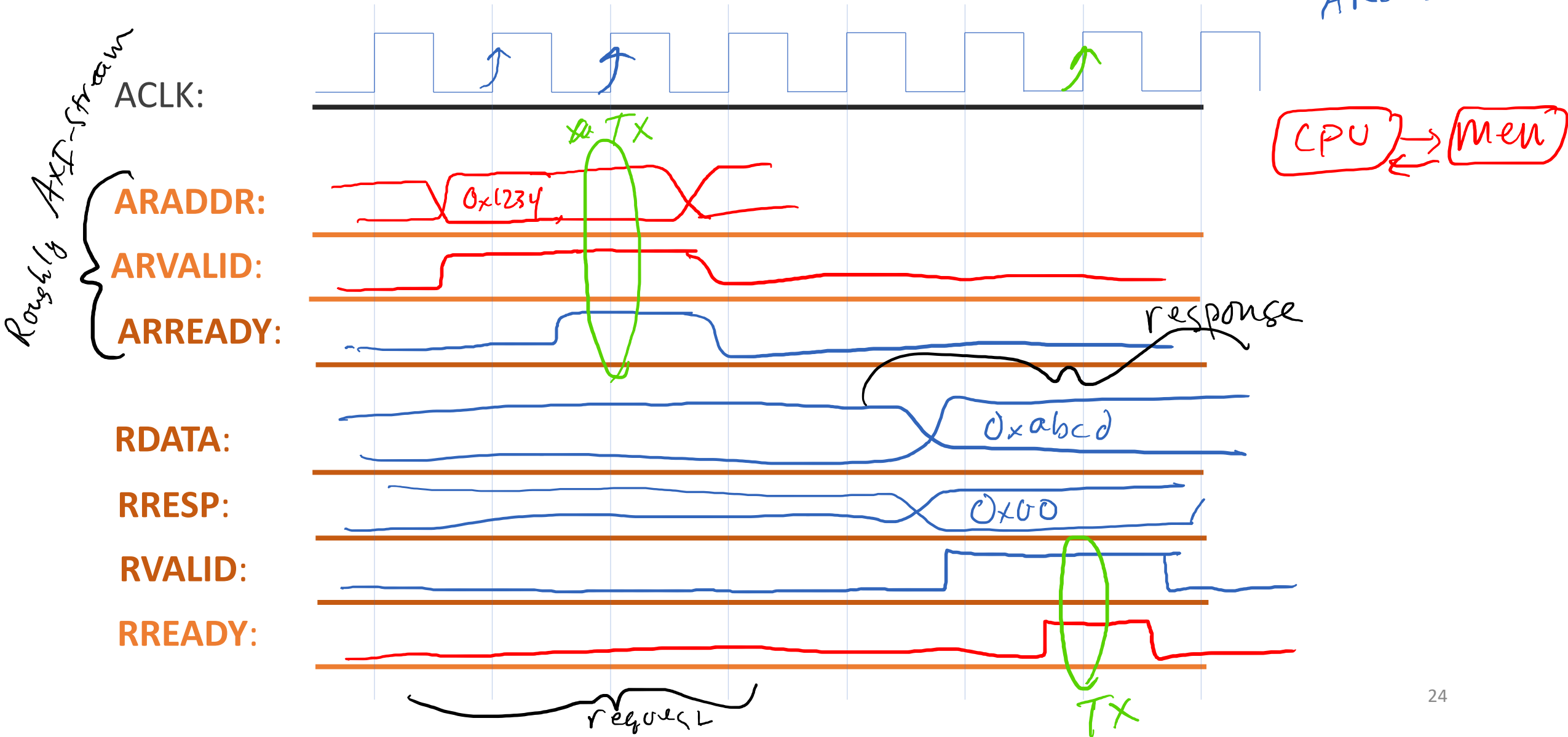
- Mostly used to send error codes back to CPU
- We'll always just use 0b00

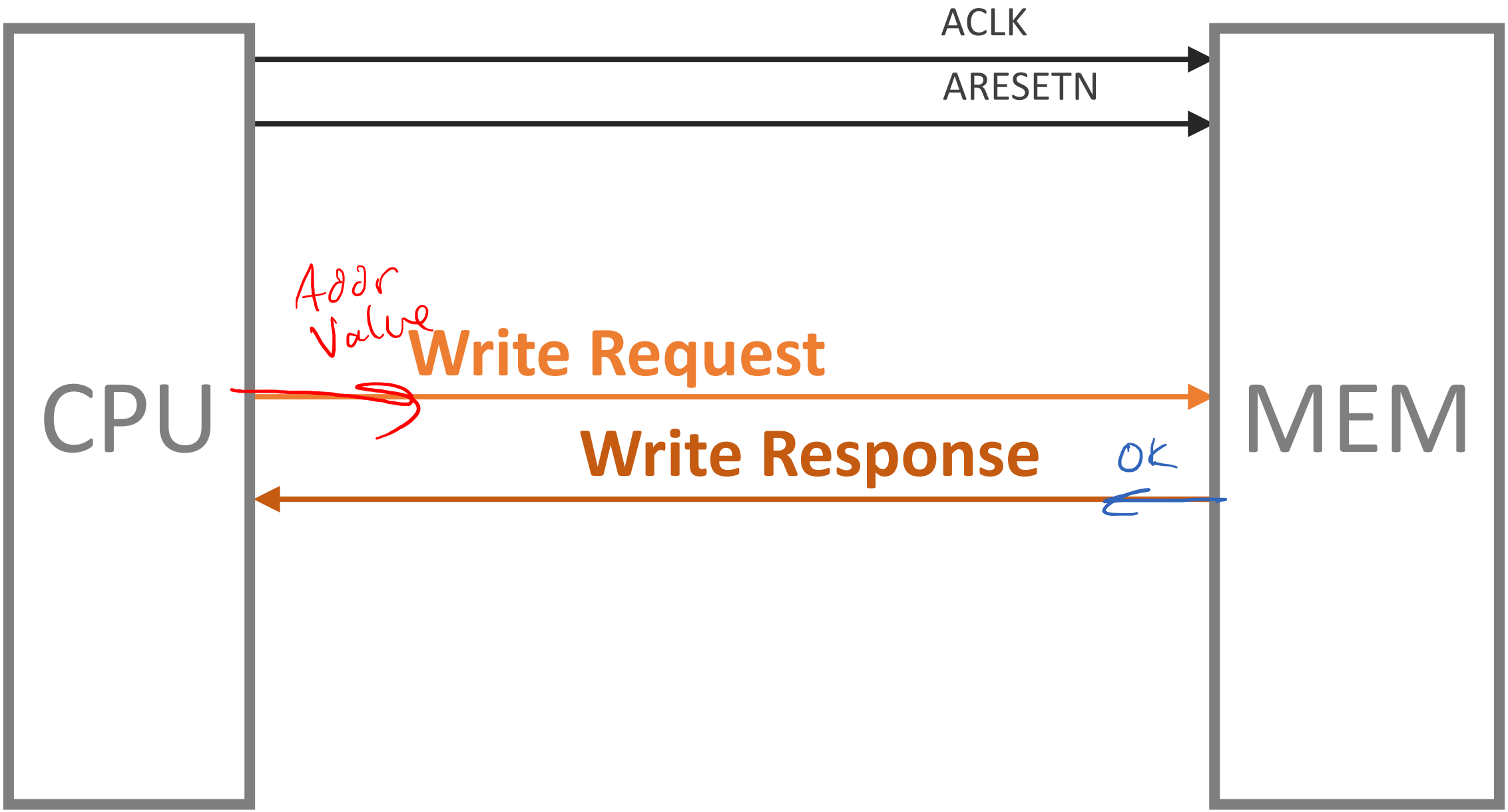
Load 0x1234, response: 0xabcd

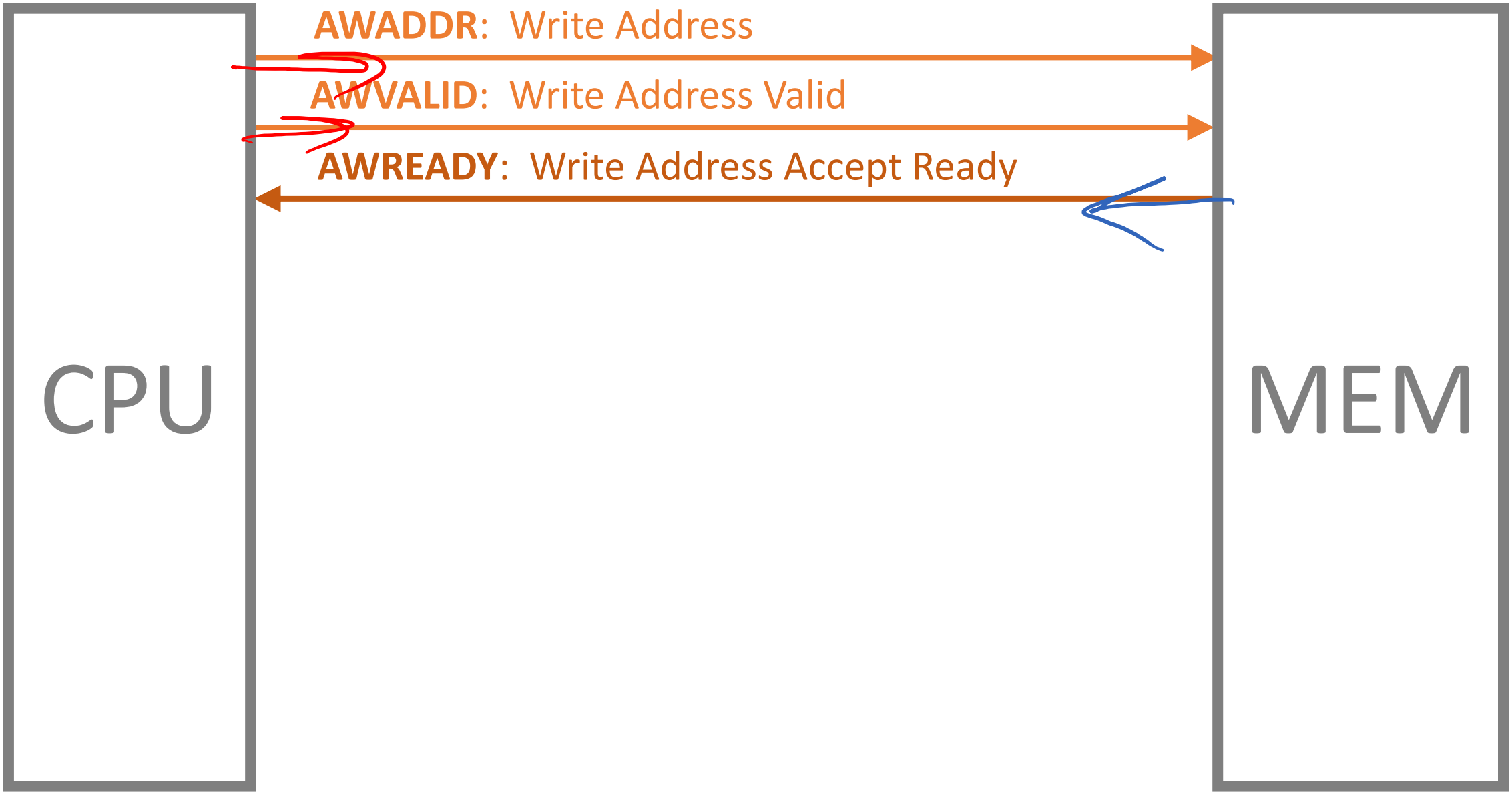
CPU

mem

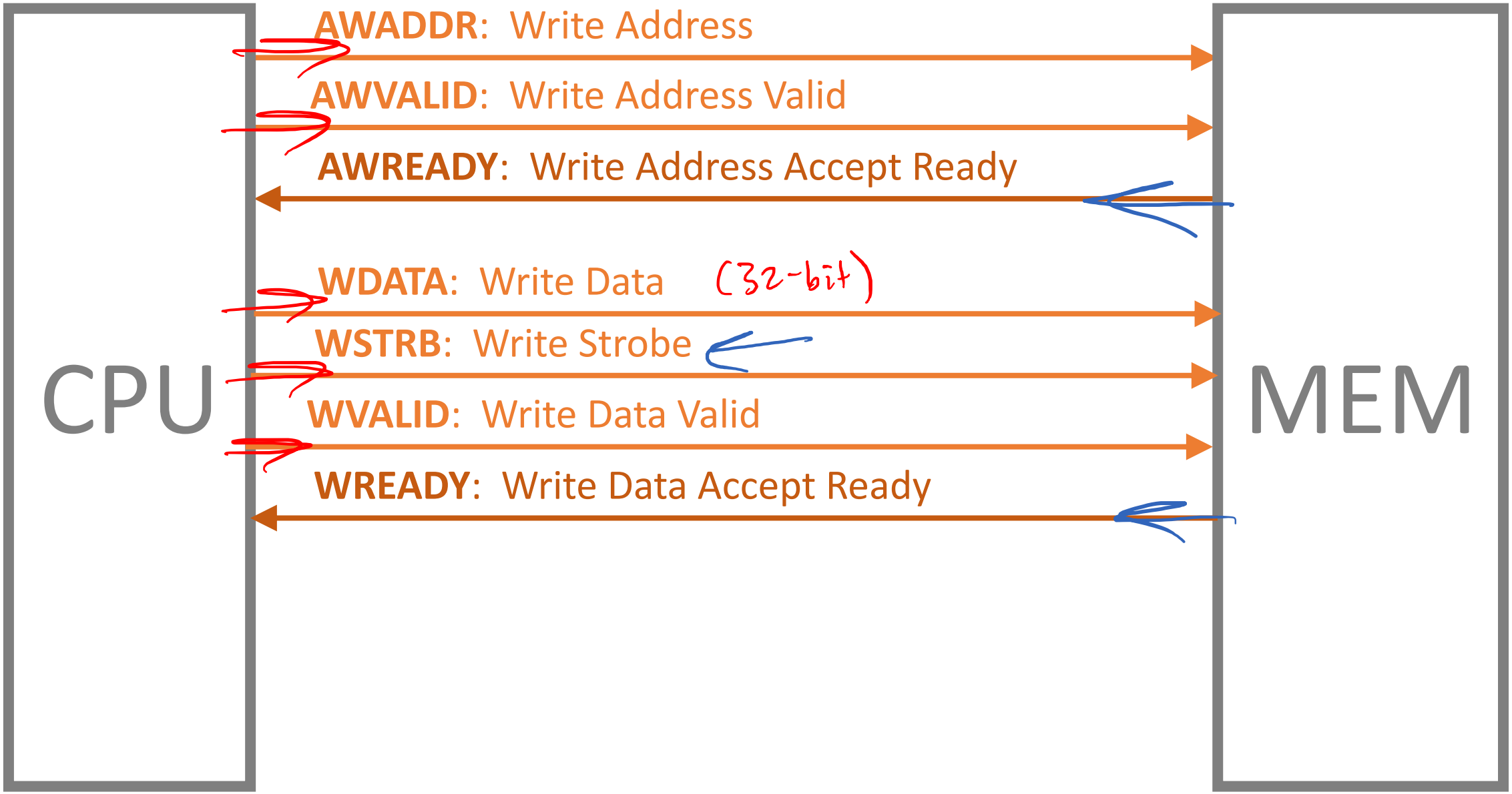
assume ARESETN = 1




ACLK
ARESETN
Addr
Value
Write Request
CPU
MEM
Write Response
OK



ACLK and ARESETN not shown



ACLK and ARESETN not shown

Q: How do you send a 1-byte (8-bit) value on a 32-bit bus?

- A: WSTB: Write Strobe

What is WSTRB?

The **WSTRB[n:0]** signals when **HIGH**, specify the byte lanes of the data bus that contain valid information. There is one write strobe for each eight bits of the write data bus, therefore **WSTRB[n]** corresponds to **WDATA[(8n)+7: (8n)]**

What is WSTRB here?

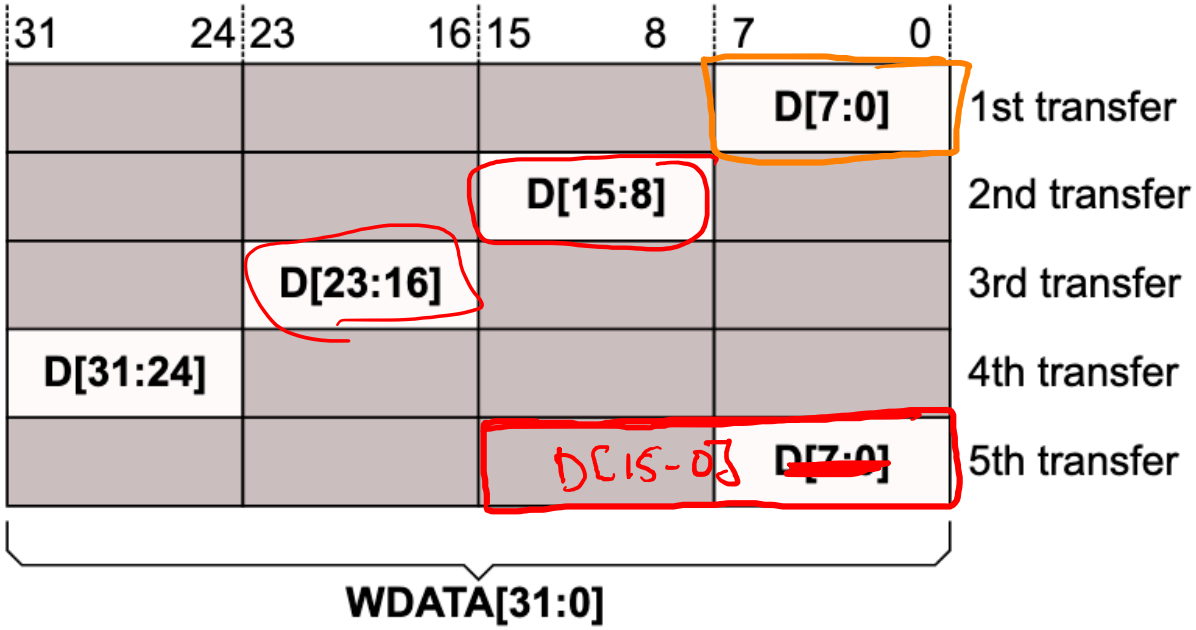


Figure A3-8 Narrow transfer example with 8-bit transfers

MSB LSB

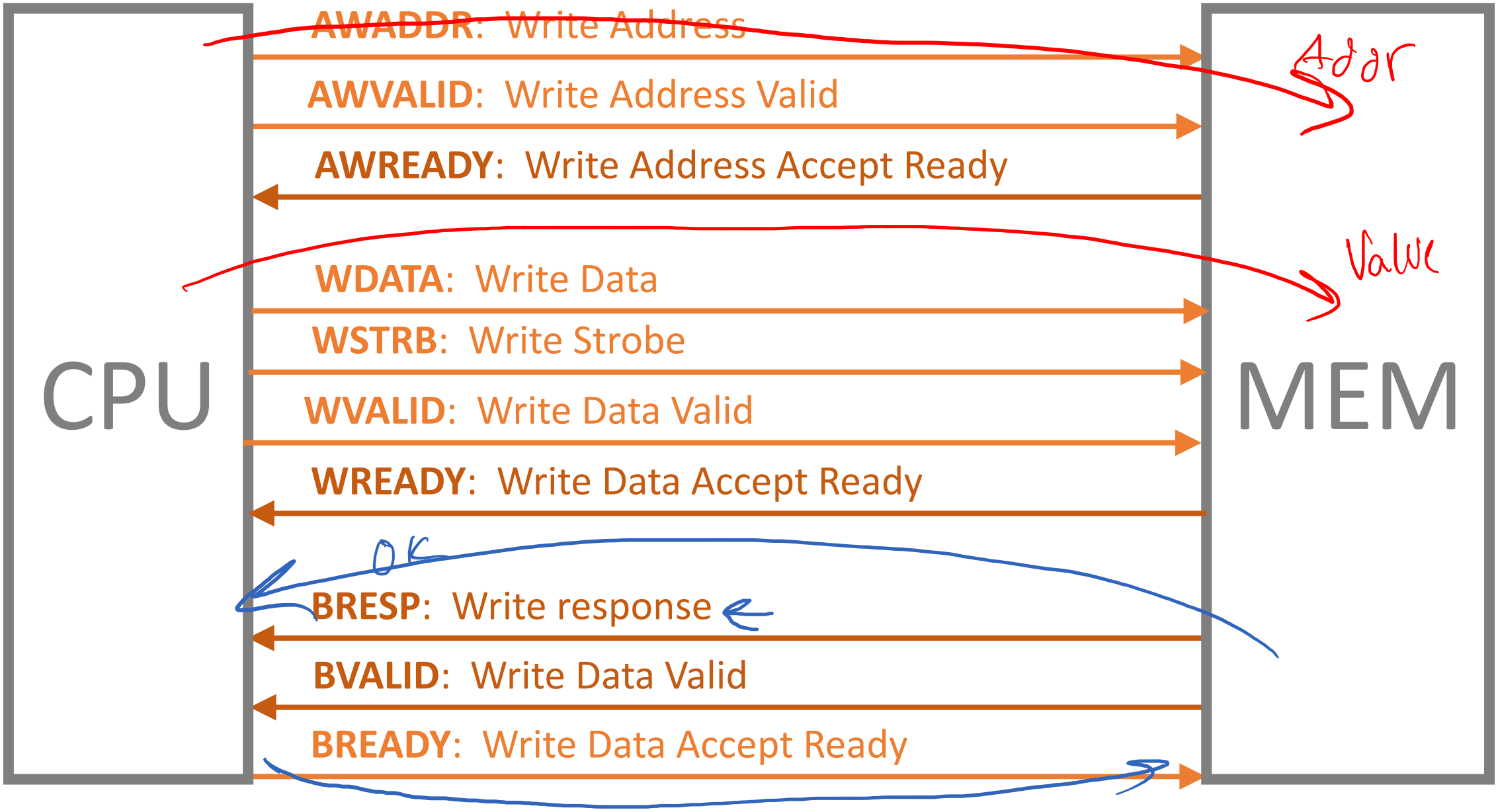
WSTRB = 0001 = 0x1

= 0010 = 0x2

= 0100 = 0x4

= 1000 = 0x8

= 0011 = 0x3



ACLK and ARESETN not shown

BRESP is just like RRESP

Table A3-4 RRESP and BRESP encoding

RRESP[1:0] BRESP[1:0]	Response
0b00	OKAY
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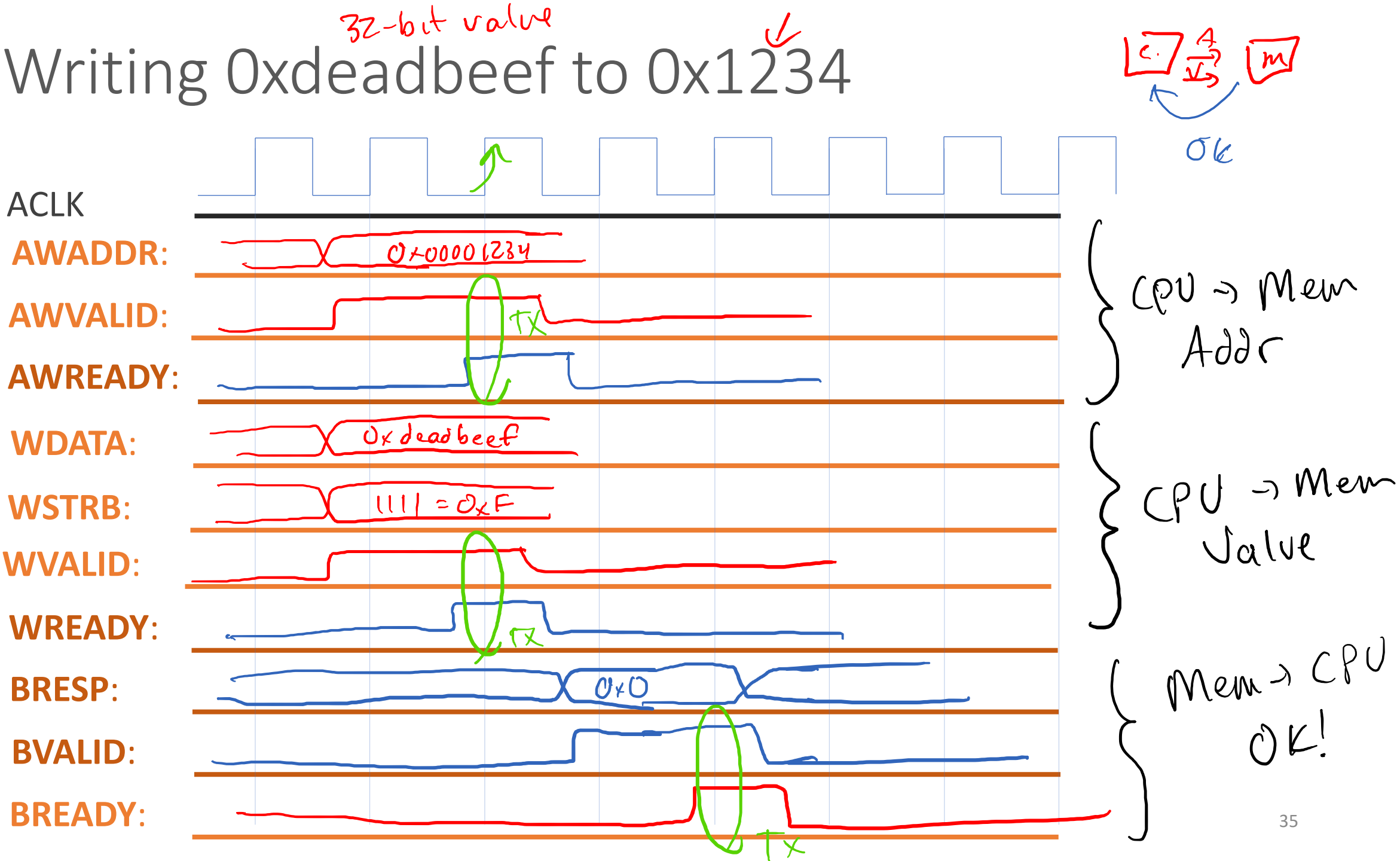
0b10	SLVERR
0b11	DECERR

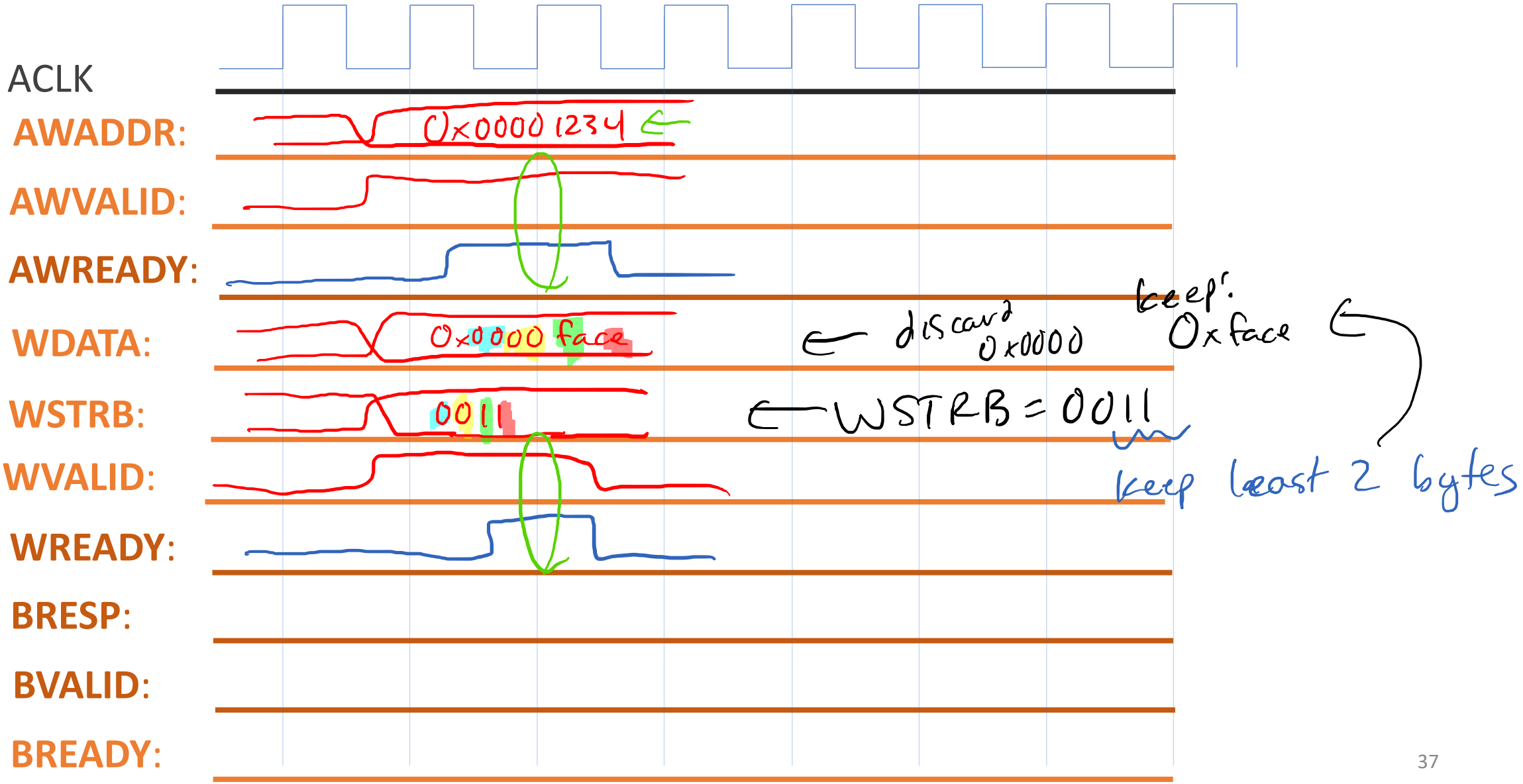
- Mostly used to send error codes back to CPU
- We'll always just use 0b00

Writing 0xdeadbeef to 0x1234



Writing 0xface to 0x1234

16-bit



Next Time

- High-Performance Busses

Monday!

References

- https://www.youtube.com/watch?v=okiTzvihHRA
- https://web.eecs.umich.edu/~prabal/teaching/eecs373/
- https://en.wikipedia.org/wiki/File:Computer_system_bus.svg
- https://www.realdigital.org/doc/a9fee931f7a172423e1ba73f66ca4081

- AMBA® AXI™ and ACE™ ProtocolSpecification

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